CTF-SSCL: CNN-Transformer for Few-Shot Hyperspectral Image Classification Assisted by Semisupervised Contrastive Learning

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***Abstract*— Few-shot learning (FSL) has rapidly advanced in the hyperspectral image classification (HSIC), potentially reducing the need for laborious and expensive labeled data collection. Due to the limited receptive field, the convolutional neural network (CNN) struggles to capture long-range dependencies for extracting global features. Additionally, the transformer focuses on global correlation while overlooking the effective representation of local spatial and spectral features. Moreover, contrastive learning (CL) has emerged as a powerful technique for improving consistency across different augmented views of samples of the same category. To this end, we devise a novel CNN-Transformer for few-shot HSIC assisted by semisupervised contrastive learning, named CTF-SSCL, to boost the classification performance. Specifically, the cascaded CNN-Transformer incorporates a lightweight spatial–spectral interactive convolution module (LSSICM) and a multiscale transformer (MSFormer) to exploit local features from submaps and global information from the entire patch. Subsequently, the semisupervised contrastive loss, comprising unsupervised and supervised components, serves as an auxiliary to optimize the model with the classification loss. Wherein, recognizing the unified spectral–spatial information in HSI, we propose a spectral feature shift strategy (SFSS) to create sample pairs for the unsupervised CL, utilizing unsupervised contrastive loss among groups of samples with identical labels. Extensive experiments on four standard benchmarks demonstrate the effectiveness of the proposed CTF-SSCL with varying amounts of labeled samples. The code will be available online at https://github.com/B-Xi/CTF-SSCL.**

***Index Terms*— Few-shot learning (FSL), lightweight, multiscale transformer (MSFormer), semisupervised contrastive learning (SSCL).**

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I. Introduction

HYPERSPECTRAL image (HSI) captures rich spectral information of land cover across hundreds of continuous channels [1], [2], [3], [4]. With this extremely high spectral resolution, HSI can distinguish different ground objects by their spectrum differences [5], [6], [7]. Based on the characteristics, HSI classification (HSIC) is capable of categorizing pixels into various classes [8], [9], [10], [11], which has been widely used in urban planning, intelligent agriculture, and environmental monitoring, to name a few [12], [13], [14].

During the past few decades, classical machine learning methods such as support vector machine (SVM) [15], extreme learning machine (ELM) [16], random forest (RF) [17], representation learning [18], and collaborative learning [19] have extensively promoted the development of HSIC. Although these methods exhibit the benefits of well-established algorithms, ease of understanding, and intuitive implementation, they predominantly focus on shallow-level features such as spectral information or handcrafted features [20], [21].

With the rapid advancement of computer vision and high-performance computing, deep learning (DL) methods have emerged, enhancing HSIC by using hierarchical frameworks to extract robust deep features in an end-to-end manner [22], [23], [24], [25], [26]. Convolutional neural networks (CNNs) effectively integrate high-dimensional hyperspectral data through 2-D [27], 3-D [28], and hybrid convolutions [29]. Additionally, lightweight convolution modules, such as the depthwise separable convolution (DSC) in MobileNet [30], [31], have been introduced to reduce model parameters. Despite these innovations, CNNs are limited by their local receptive fields and complex convolutions, which constrain their ability to capture global spatial information in HSI and highlight different spectral–spatial features effectively [32].

Recently, attention mechanisms have been extensively explored to enhance the performance of HSIC by emphasizing valuable features and reducing the impact of noisy

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information [33]. For example, in [34], a method based on the traditional CNN model was proposed, incorporating attention modules to assist networks in focusing on more discriminative channels or positions. Additionally, Shen et al. [35] integrated nonlocal blocks as attention mechanisms within the CNN architecture to capture nonlocal dependencies in hyperspectral data. Transformer, a self-attention mechanism that is well-designed to process and analyze sequential data (or time series), has demonstrated a strong capability in modeling long-distance dependencies [36]. Recent studies based on transformers have shown outstanding performance in HSIC. For instance, Hong et al. [37] developed a SpectralFormer that effectively learns detailed spectral representations and transfers memory-like components across different network layers. To address the issue of over-dispersion in features extracted by multihead self-attention (MHSA), a group-aware hierarchical transformer (GAHT) for HSIC was introduced to focus attention on local spatial–spectral contexts [38]. Moreover, Jiang et al. [36] proposed GraphGST model, which aggregates local-to-global correlations by integrating graph structural information into the transformer architecture and improves the classification performance.

However, the challenge arises when there are limited annotated samples, resulting in a significant decrease in classification accuracy. In practical terms, acquiring and annotating a large amount of labeled data are both time-consuming and costly, especially given the numerous classes in HSI [39], [40]. Few-shot learning (FSL), which is essentially a meta-learning approach, is gaining traction in HSIC due to its ability to acquire general knowledge from extensive learning tasks and apply it to swiftly address new challenges [41]. Specifically, FSL utilizes K-labeled training samples of each class to train an N-class classification model (referred to as N-way K-shot). This approach makes it easier to distinguish novel-class samples in the embedding space by employing a linear classifier based on distance measures [42]. For example, Li et al. [43] recently introduced a deep cross-domain few-shot learning (DCFSL) framework, which considers the domain shift between the base and novel classes and utilizes a popular deep residual 3-D CNN (3D-ResNet) for embedded feature learning, achieving state-of-the-art classification results. Additionally, Wang et al. [44] simultaneously addressed local spatial alignment and spectral alignment to transfer knowledge from the source domain to the target domain. They balanced domain-invariant and domain-specific features using a weakly parameter shared mechanism. Moreover, Liu et al. [45] proposed a refined prototypical contrastive learning network for FSL (RPCL-FSL), which incorporates supervised contrastive learning (CL) and FSL into an end-to-end network to perform small-sample HSIC. Furthermore, a compact and innovative transformer network, spectral coordinate transformer (SCFormer), was developed in [46] to explore spectral coordinate priors, reducing spectral position disturbance caused by convolution operations and enhancing the robustness of the method.

From the above methods, the effectiveness of CNNs in processing local features and transformers in capturing global receptive fields has been confirmed, highlighting the strengths of each model. Recent research has shown that integrating CNNs and Transformers leads to models with robust feature extraction capabilities. For instance, Peng et al. [47] developed a few-shot approach based on a convolutional transformer, which enhances feature embedding efficiency by combining the outputs of a CNN and a vision transformer (ViT). Similarly, Bai et al. [48] proposed a novel method that integrates a generative adversarial network (GAN), transformer encoder, and convolution block within a unified framework, eliminating the need for an external dataset. However, these approaches encounter challenges in handling a large number of parameters when dealing with complex hyperspectral data. Therefore, we are concerned about designing an efficient model with fewer parameters for FSL that leverages both CNN and transformer architectures.

Moreover, achieving accurate classification with only a limited number of labeled samples remains a difficult task using the FSL strategy (N-way K-shot) [49]. To address this challenge, instead of focusing on learning individual sample features, we notice that CL can capture the similarity and dissimilarity distribution properties of sample pairs in the latent feature space [50]. This approach helps reduce the requirement for extensive training samples while providing an effective network training mechanism for learning highly distinctive features. Additionally, CL can be applied to both labeled and unlabeled data in supervised and unsupervised manners, respectively [51], [52], [53].

In light of these considerations, we propose a novel cascaded CNN-Transformer network for few-shot HSIC assisted by a semisupervised contrastive learning (CTF-SSCL) scheme. First, to facilitate local spatial–spectral interaction in a lightweight manner, we integrate efficient convolutional operators, such as groupwise convolution (GWC) and depthwise convolution (DWC), into a comprehensive convolution module. Second, we propose a multiscale transformer (MSFormer) that includes local–global self-attention (LGSA) and cross-group self-attention (CGSA) mechanisms to capture dual-dimensional information in HSI. Lastly, to enhance discriminative feature embedding, we extend CL to incorporate both supervised and unsupervised perspectives, leading to the design of a semisupervised contrastive loss function. Notably, to generate sample pairs for unsupervised CL, we develop a specific data augmentation technique for HSI based on random perturbation operations.

The main contributions of the proposed frameworks can be summarized as follows.

1) We propose a novel CNN-Transformer network comprising a lightweight spatial–spectral interactive convolution module (LSSICM) and an MSFormer to accommodate local features and long-range information.
2) We devise a semisupervised loss function based on CL principles to obtain more discriminative features. Specifically, we calculate the unsupervised contrastive loss using groups rather than the traditional manner of involving individual samples.
3) Regarding the merged spectral–spatial information in the HSI, we propose a spectral feature shift strategy

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(SFSS) based on random perturbation along the spectral dimension, generating paired samples for unsupervised CL.

The remainder of this article is organized as follows. Section II briefly introduces the related works. Section III describes our proposed method in detail. Section IV carries out abundant experiments and analyzes the classification results. Section V draws the conclusion.

II. Related Works

A. FSL for HSIC

FSL is a highly effective learning approach that enables rapid generalization to new tasks with limited supervisory experience and prior knowledge [54]. Meta-learning, as one of the prominent solutions in FSL, mimics the biological process of leveraging existing knowledge to quickly adapt to learning new and unseen tasks. Specifically, in an N-way K-shot classification task, the objective is to utilize K-labeled training data for each class to develop an N-class classification model, typically with a small value for K.

In the field of HSIC, meta-learning-based approaches have also been studied profoundly. For instance, Liu et al. [42] introduced the deep few-shot learning (DFSL) model, which employs meta-learning to train feature extraction models using accurately labeled public hyperspectral datasets and subsequently classifies the deep spatial–spectral meta features of new scenes using SVM or nearest neighbor classifiers. Rao et al. [55] proposed the spatial–spectral relation network, which combines labeled support and query samples to augment training data and utilizes a relation network architecture to capture significant similarities between samples. To enhance performance, Xi et al. [56] presented a class-covariance-based metric space for FSL (CMFSL) in HSIC. Additionally, Zhang et al. [57] developed the cross-domain similarity-aware block based on graph information aggregation for FSL (Gia-CFSL) to address domain shifts between HSI scenes. Furthermore, the adaptive domain-adversarial FSL (ADAFSL) method proposed in [58] employs an adaptive strategy that assigns varying weights to conditional adversarial losses for different classes based on their respective discrepancies, thereby achieving alignment in the global conditional distribution.

B. Transformer Mechanism for HSIC

In HSIC tasks, the training samples typically have a cube format. However, due to the limited size of the receptive field compared to the patch size, there is a loss of global context for the entire cube. To overcome this limitation, the ViT incorporates an MHSA mechanism [59] to address global dependencies within the image. The architecture of the MHSA and the contained scaled dot-product operation is illustrated in Fig. 1. Initially, the spatial dimension of the input patch $\mathbf{X}_0 \in \mathbb{R}^{h\times w\times d}$ is squeezed to form a 2-D input matrix $\mathbf{X}$ with dimensions of $s \times d$, where s equals $h \times w$. Subsequently, $\mathbf{X}$ is transformed into three distinct matrices including query $\mathbf{Q} \in \mathbb{R}^{s\times d}$, key $\mathbf{K} \in \mathbb{R}^{s\times d}$, and value $\mathbf{V} \in \mathbb{R}^{s\times d}$ through the linear transformations

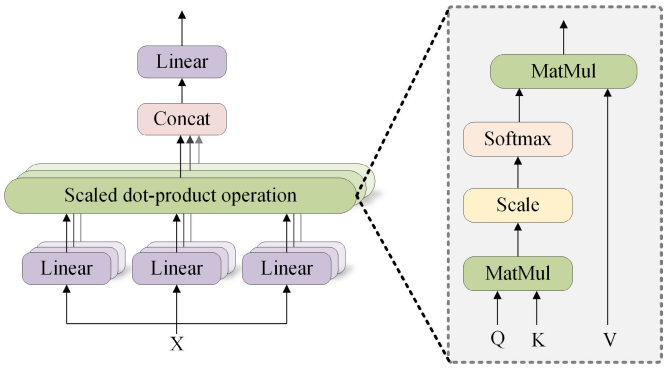


Fig. 1. MHSA mechanism and the contained scaled dot-product operation.

$$\mathbf{Q} = \mathbf{X}\mathbf{W}^{Q}, \quad \mathbf{K} = \mathbf{X}\mathbf{W}^{K}, \quad \mathbf{V} = \mathbf{X}\mathbf{W}^{V}. \tag{1}$$

Here, $\mathbf{W}^{Q} \in \mathbb{R}^{d\times d}$, $\mathbf{W}^{K} \in \mathbb{R}^{d\times d}$, and $\mathbf{W}^{V} \in \mathbb{R}^{d\times d}$.

Following this, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are divided into n heads along the dimensions:

$$\mathbf{Q} = \{\mathbf{Q}_1, \mathbf{Q}_2, \ldots, \mathbf{Q}_i, \ldots, \mathbf{Q}_n\} \tag{2}$$

$$\mathbf{K} = \{\mathbf{K}_1, \mathbf{K}_2, \ldots, \mathbf{K}_i, \ldots, \mathbf{K}_n\} \tag{3}$$

$$\mathbf{V} = \{\mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_i, \ldots, \mathbf{V}_n\} \tag{4}$$

where $\mathbf{Q}_i$, $\mathbf{K}_i$, and $\mathbf{V}_i \in \mathbb{R}^{s\times(d/n)}$. The ith attention score is calculated using a scaled dot-product operation

$$\mathbf{Z}_i = \text{attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) = \text{softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^{\mathrm{T}}}{\sqrt{d}}\right)\mathbf{V}_i. \tag{5}$$

The final output of MHSA is obtained through a projection after concatenating all $\mathbf{Z}_i \in \mathbb{R}^{s\times(d/n)}$ as follows:

$$\mathbf{O} = \text{Concat}(\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_i, \ldots, \mathbf{Z}_n)\mathbf{W}^{O}. \tag{6}$$

Here, the projection matrix $\mathbf{W}^{O} \in \mathbb{R}^{d\times d}$ and $\mathbf{O} \in \mathbb{R}^{s\times d}$.

Known for its ability to establish long-range dependencies and extract crucial features, the transformer has demonstrated exceptional performance in processing HSIC. The initial work in HSIC, as referenced in [60], involved the direct adoption of bidirectional encoder representations from transformers (BERTs) on flattened input patches. After that, various transformer-based methodologies have been introduced. For instance, Qiu et al. [61] utilized multiscale convolution and cross-channel fusion to facilitate information interaction between channels. Additionally, a transpose multihead self-attention (TMHSA) was developed to extract global features. Furthermore, the GAHT mentioned earlier [38] introduces a novel grouped pixel embedding module for HSIC, emphasizing local relationships within spectral channels and achieving promising classification performance.

III. Proposed Method

The architecture of the proposed method is illustrated in Fig. 2. Following the meta-learning, we utilize labeled samples $\mathbf{X}^{L}$ to construct both support and query datasets. In contrast, unlabeled samples $\mathbf{X}^{U}$ are utilized to generate pairs from different views. Additionally, we establish a feature

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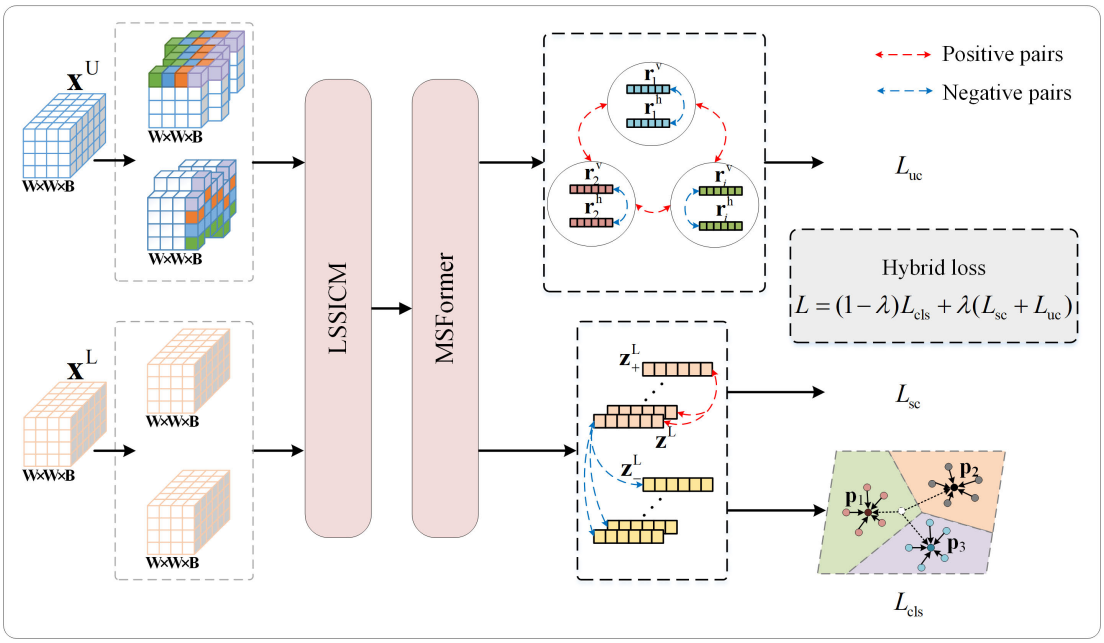


Fig. 2. Architecture of the proposed CTF-SSCL for HSIC. The feature extractor is comprised of the LSSICM and MSFormer. Concretely, a hybrid loss function including classification loss L_{cls}, supervised contrastive loss L_{sc}, and unsupervised contrastive loss L_{uc} is introduced to train the network.

extractor consisting of LSSICM and MSFormer, along with a hybrid loss function incorporating unsupervised contrastive loss L_{uc}, supervised contrastive loss L_{sc}, and classification loss L_{cls}. Specifically, L_{uc} focuses on ensuring class consistency between two groups with the same label (i.e., $\mathbf{r}_i^h$ and $\mathbf{r}_i^v$) in two pathways. L_{sc} aims to reduce the distance between features of the same category from $\mathbf{X}^L$. After the training process, unlabeled samples are fed into the trained embedding network to produce features for classification purposes.

A. Lightweight Spatial–Spectral Interactive Convolution Module

The presented LSSICM, a local spectral–spatial feature extractor illustrated in Fig. 3, is composed of GWC and cross-channel fusion module (CCFM). This module primarily utilizes two types of effective convolutions: GWC and DWC. Additionally, a shortcut is implemented in the LSSICM, akin to the functionality of ResNet [62], to effectively mitigate overfitting.

1) Groupwise Convolution: In contrast to standard convolution, GWC performs distinct operations. The comparison between standard convolution and GWC is depicted in Fig. 4(a) and (b). Instead of directly applying convolutions with a specific kernel size, GWC first divides the input feature maps $\mathbf{X}_{in}$ into g nonoverlapped and equally sized groups along the channel dimension

$$\mathbf{X}_{in} = \{\mathbf{X}_{in1}, \mathbf{X}_{in2}, \ldots, \mathbf{X}_{ing}\}. \tag{7}$$

Subsequently, a standard convolution operation is independently executed on each group, and the outputs of these groups are concatenated sequentially.

The GWC employed here is devoted to grasping grouped spatial–spectral information with fewer parameters than the standard convolution. Specifically, if the shape of the input feature map $\mathbf{X}_{in}$ is $C_{in} \times H \times W$ and the resulting output feature map $\mathbf{X}_{out}$ has dimensions of $C_{out} \times H' \times W'$, standard convolution would necessitate C_{out} convolution kernels with a size of $C_{in} \times h \times w$. In contrast, the GWC utilizes the same C_{out} number of standard convolution kernels, but with a reduced size of $(C_{in}/g) \times h \times w$, divided

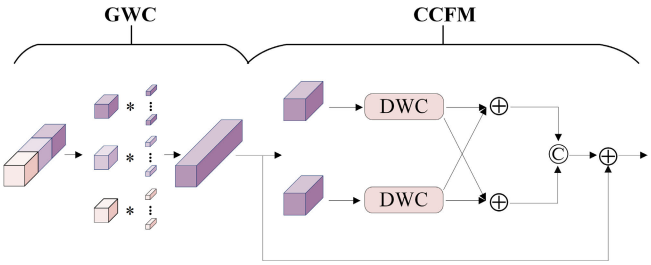


Fig. 3. Detailed architecture of the proposed LSSICM.

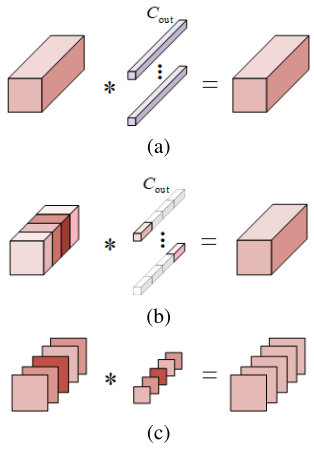


Fig. 4. (a) Standard convolution. (b) GWC. (c) DWC.

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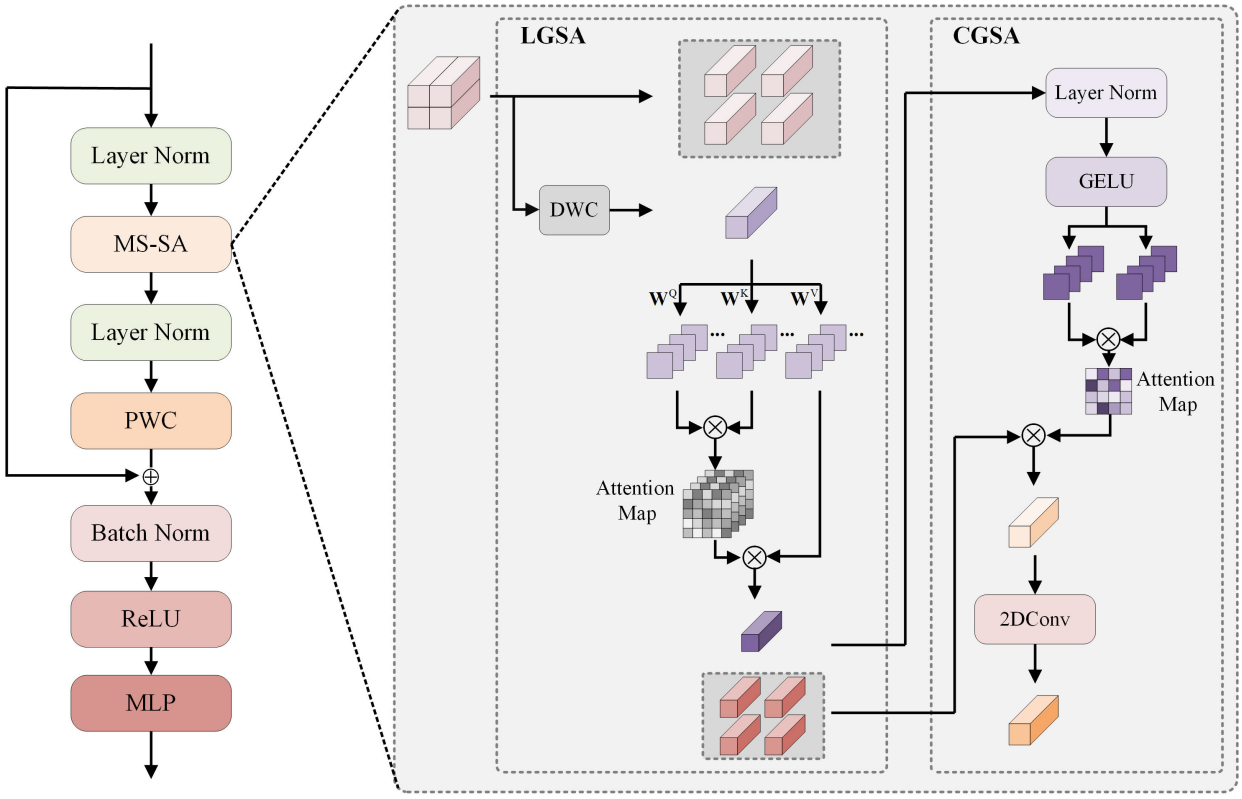


Fig. 5. Architecture of the proposed MSFormer. The MS-SA comprises an LGSA and a CGSA.

into g groups. Consequently, by employing GWC, the parameter count decreases from $C_{\text{out}} \times C_{\text{in}} \times h \times w$ to $C_{\text{out}} \times C_{\text{in}}/g \times h \times w$.

2) Cross-Channel Fusion Module: To leverage the rich spectral information, we devise a CCFM with efficient computations. For the input $\mathbf{X}_{\text{in}}$, we first partition the channels into two equal parts, denoted as $\mathbf{X}_1$ and $\mathbf{X}_2$. In order to capture spatial relationships and aggregate local spatial information, we utilize identical DWC with shared parameters on $\mathbf{X}_1$ and $\mathbf{X}_2$. As depicted in Fig. 4(c), the DWC is a lightweight operation that performs convolution independently on each input channel. Following the channelwise execution of DWC, we obtain spatial global features $\hat{\mathbf{X}}_1$ and $\hat{\mathbf{X}}_2$. Then, to facilitate information interaction between channels, we merge these spatial global features across different channels as follows:

$$\mathbf{X}_1' = \hat{\mathbf{X}}_1 + \alpha\hat{\mathbf{X}}_2 \tag{8}$$

$$\mathbf{X}_2' = \hat{\mathbf{X}}_2 + \alpha\hat{\mathbf{X}}_1 \tag{9}$$

where α is a hyperparameter less than 1. Then, we concatenate $\mathbf{X}_1'$ and $\mathbf{X}_2'$ to acquire the output $\mathbf{X}' \in \mathbb{R}^{C_{\text{out}} \times H' \times W'}$

$$\mathbf{X}' = \text{Concat}\left(\mathbf{X}_1', \mathbf{X}_2'\right). \tag{10}$$

It is important to note that the CCFM not only emphasizes the local spatial features through DWC but also enables information exchange between channels via cross-channel fusion. In comparison to the conventional DSC with similar functionalities, the CCFM occupies fewer parameters. Specifically, the former requires $C_{\text{out}} \times h \times w + C_{\text{out}} \times C_{\text{out}}$ parameters, while the parameter count of the latter is $C_{\text{out}}/2 \times h \times w$. Consequently, the parameter ratio between the DSC and CCFM is calculated as

$$\frac{P_{\text{DSC}}}{P_{\text{CCFM}}} = \frac{C_{\text{out}} \times h \times w + C_{\text{out}} \times C_{\text{out}}}{\frac{C_{\text{out}}}{2} \times h \times w} = 2 + \frac{2C_{\text{out}}}{hw}. \tag{11}$$

It is noted from (11) that DSC has more than twice the parameter count of the CCFM.

B. Multiscale Transformer

As traditional ViT often lacks local perception, we design the MSFormer combining local and global considerations, which is illustrated in Fig. 5. The principal component of the framework is multiscale self-attention (MS-SA), which incorporates an LGSA and a CGSA to emphasize the efficient extraction of global and local spatial features.

1) Local–Global Self-Attention: In contrast to applying MHSA on all tokens within a patch, we initially group patches along the spatial dimension to obtain local tokens. Specifically, the spatial dimensions of the input feature maps are partitioned into p equally sized submaps. Subsequently, a "local token" is generated within each group to serve as the core feature representation, facilitating a global representation of the features. Additionally, we utilize the DWC on the entire patch to squeeze the spatial dimensions and create a global map of the same size as the submaps, serving as a global token for a comprehensive representation of the patch. Following this, MHSA is performed on all feature tokens, encompassing both local and global tokens. To be more

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specific, the implementation of the LGSA is represented as

$$\mathrm{LGSA}(j) = \mathrm{MHSA}(\mathbf{Q}_j, \mathbf{K}_j, \mathbf{V}_j) \tag{12}$$

where j denotes the jth submap or the global map in the LGSA. Specifically, assuming the input feature map size is $C \times \hat{H} \times \hat{W}$, with each group having spatial dimensions of $M \times M$, the total number of group tokens is $N = (\hat{H}\hat{W}/M^2)$ and $\mathbf{Q}_j, \mathbf{K}_j, \mathbf{V}_j \in \mathbb{R}^{M^2 \times C}$.

2) Cross-Group Self-Attention: In this phase, we leverage the output from LGSA to build local–global connections. The objective of the global token is to allocate global weights to each feature submap. After passing through the layer normalization (LN) layer and Gaussian error linear unit (GELU) activation function denoted as $\mathrm{GELU} = (1/2)(1 + \int_0^{(x/\sqrt{2})} e^{-t^2} dt)$, the attention weight matrix is generated. Subsequently, attention calculations are executed between the attention weight matrix and the subfeature maps, facilitating the fusion of global information. The process of the CGSA is delineated as follows:

$$\mathbf{Q}_\Lambda = \mathrm{LN}(\mathrm{GELU}(\mathbf{\Lambda}))\mathbf{W}^{Q_\Lambda} \tag{13}$$

$$\mathbf{K}_\Lambda = \mathrm{LN}(\mathrm{GELU}(\mathbf{\Lambda}))\mathbf{W}^{K_\Lambda} \tag{14}$$

$$\mathrm{CGSA}(\mathbf{Q}_\Lambda, \mathbf{K}_\Lambda, \mathbf{L}) = \mathrm{MHSA}(\mathbf{Q}_\Lambda, \mathbf{K}_\Lambda, \mathbf{L}) \tag{15}$$

where $\mathbf{\Lambda} \in \mathbb{R}^{N \times C}$ signifies the global token and $\mathbf{W}^{Q_\Lambda}, \mathbf{W}^{K_\Lambda} \in \mathbb{R}^{C \times C}$. Within CGSA, attention functions as a standard self-attention operator, functioning across the entire feature maps in units of subfeature maps denoted as $\mathbf{L}$ (i.e., local tokens).

Overall, the integration of LGSA and CGSA enables the MSFormer to extract both local and global spatial features, addressing the limitation of standard MHSA, which exhibits strong global dependence but lacks local dependence.

C. SSCL for HSIC

As mentioned, unsupervised CL aims to learn representations by maximizing agreement between different augmented views of the same unlabeled sample. On the other hand, supervised CL works toward bringing samples with the same category closer to each other in the embedding space. To effectively train the proposed end-to-end framework, we develop a hybrid loss function that combines the unsupervised contrastive loss, supervised contrastive loss, and classification loss.

1) Unsupervised Contrastive Loss: A contrastive objective applied between instances may inadvertently lead to the separation of different instances from the same class. To mitigate this issue, we group unlabeled samples with the same class and introduce an unsupervised contrastive loss among them. To handle the merged spectral–spatial information in the HSI, we introduce an SFSS to generate paired samples from each unlabeled sample $\mathbf{z}^U$. This strategy involves a bidirectional movement of μ randomly selected spectral bands along the horizontal and vertical dimensions of the feature maps, significantly enhancing the diversity of input features and improving the model's robustness.

Assuming y_i^h and y_i^v represent the pseudo-labels of the sample $\mathbf{z}_i^U$ in two pathways, respectively, samples with the same pseudo-label in a batch can form a group in each pathway. The average of these samples, denoted as $\mathbf{r}_l^p$, serves as the representation of the specific group and is formulated as follows:

$$\mathbf{r}_l^p = \frac{\sum_{i=1}^{B} 1_{\{\hat{\mathbf{y}}_i^p\}} \mathbf{z}_i^p}{T} \tag{16}$$

where 1 is an indicator function representing samples with a pseudo-label equal to $l \in Y$ in each pathway $p \in \{h, v\}$. T indicates the number of such samples in the batch, while B denotes the batch size.

To promote class consistency among two groups that share the same label in different pathways, our focus is on generating similar representations for these groups in the feature space. Considering $(\mathbf{r}_i^h, \mathbf{r}_i^v)$ as positive pairs and $(\mathbf{r}_l^h, \mathbf{r}_m^p)$ as negative pairs with $p \in \{h, v\}$ and $m \in Y \backslash l$, ensuring that the constituent groups differ in either pathway, the formulation of unsupervised contrastive loss is as follows:

$$L_{\mathrm{uc}}(\mathbf{r}_l^h, \mathbf{r}_l^v) = -\log \frac{h\left(\mathbf{r}_l^h, \mathbf{r}_l^v\right)}{h\left(\mathbf{r}_l^h, \mathbf{r}_l^v\right) + \sum\limits_{\substack{m=1 \\ p \in \{h,v\}}}^{\beta} 1_{\{m \neq l\}} h\left(\mathbf{r}_l^h, \mathbf{r}_m^p\right)} \tag{17}$$

where $h(u, v) = \exp((u^{\mathrm{T}} v / \|u\|_2 \|u\|_2)/\epsilon)$ represents the exponential of the cosine similarity and ϵ is a temperature hyperparameter. β corresponds to the class number of the unlabeled samples.

2) Supervised Contrastive Loss: Supervised CL aims to force the model to learn discriminative features for labeled samples. This is achieved by maximizing the similarity between the representations of intraclass samples while minimizing the similarity of interclass samples. Consequently, this process reduces the distance between samples of the same category and enhances the separation between samples of different classes. Given a labeled sample $\mathbf{z}^L \in D^L$ with its corresponding ground-truth label $y \in Y$, the supervised contrastive loss is computed as follows:

$$L_{\mathrm{sc}} = -\log \frac{\sum_{y=y_+} \exp\left(\mathrm{sim}\left(\mathbf{z}^L, \mathbf{z}_+^L\right)/\tau\right)}{\sum_{y=y_-} \exp\left(\mathrm{sim}(\mathbf{z}^L, \mathbf{z}_-^L)/\tau\right)} \tag{18}$$

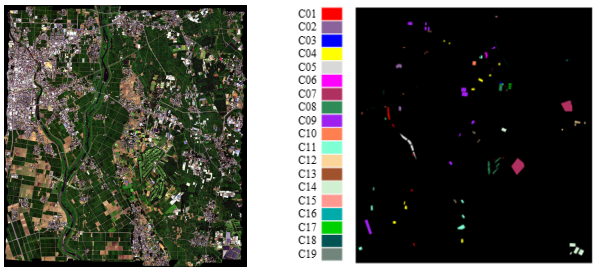
where sim denotes the cosine similarity and τ represents a temperature factor. The $\mathbf{z}_+^L$ is a representation derived from a sample belonging to the same class as $\mathbf{z}^L$, while $\mathbf{z}_-^L$ is derived from a sample with a different label.

3) Classification Loss: We employ a classification loss to directly optimize the model for the classification task. Following multiple FSL tasks for HSIC, we calculate the average support features for each category as prototypes $\mathbf{p}_i$. Then, the predicted class probability $\hat{\mathbf{y}}_s$ for query data is determined based on the Euclidean distance between the query feature and the prototype of each class. We minimize the cross-entropy loss (L_{cls}) between predicted class probability $\hat{\mathbf{y}}_s$ and the ground-truth label $\mathbf{y}_s$ as follows:

$$L_{\mathrm{cls}} = -\sum_{s=1}^{S} \mathbf{y}_s \log \hat{\mathbf{y}}_s \tag{19}$$

where S is the number of samples.

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(a) (b)

Fig. 6. Chikusei dataset. (a) False-color image. (b) Ground-truth map.

4) Loss Function: To enhance the discriminative features, we carefully design the loss function. Specifically, the classification loss and the semisupervised contrastive loss comprised of supervised and unsupervised contrastive losses are considered. The overall loss function is formulated as

$$L = (1-\lambda)L_{\text{cls}} + \lambda(L_{\text{sc}} + L_{\text{uc}}) \tag{20}$$

where λ represents the weight hyperparameter.

IV. Experimental Results and Analysis

A. Datasets and Evaluation Measures

The experimental datasets utilized in this work are categorized into two groups: the source dataset containing base classes with sufficient labeled samples, and the target datasets comprising novel classes with few-shot annotated instances. Referred to [41], [43], and [57], the Chikusei is selected as the source dataset, serving as the auxiliary training data. The Indian Pines (IP), Pavia University (UP), WHU_Hi_HanChuan (HC), and Kennedy Space Center (KSC) are chosen as the target datasets.

The Chikusei dataset was collected using the hyperspectral visible/near-infrared camera (Hyperspec-VNIR-C) over Chikusei, Japan, in 2014. Fig. 6 displays its false-color image and ground-truth map. It has a spatial size of 2517 × 2335, with up to 19 labeled classes detailed in Table I. The spatial resolution is 2.5 m per pixel, encompassing 128 spectral bands ranging from 363 to 1018 nm.

For the four target datasets, the groundtruth, labeled land-cover types, and detailed sample numbers are presented in Table II. Different colors in each dataset represent different land covers. Specifically, the IP dataset is acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor with 224 bands from 0.4 to 2.5 μm, retaining 200 bands after excluding the water absorption spectrum. The image comprises 145 × 145 pixels with a 20-m spatial resolution. The UP dataset is collected by the Reflective Optics Spectrographic Imaging System (ROSIS) airborne instrument. It comprises 610 × 340 pixels with a 1.3-m spatial resolution and 103 spectral bands ranging from 0.43 to 0.86 μm after abandoning 12 noisy channels. The HC dataset is collected by the 17-mm focal length Headwall Nano-Hyperspec imaging sensor installed on the V1 platform of the Leica Aibot X6 unmanned aerial vehicle (UAV). The UAV flight height is 250 m, and the image size is 1217 × 303 pixels, with 274 bands from 0.4 to 1.0 μm. The spatial resolution of the UAV-borne HSI is approximately 0.109 m. Notably, the dataset contains numerous shadow coverage areas due to the image being captured in the afternoon with a low solar elevation angle. The KSC dataset was collected by the NASA AVIRIS instrument in 1996. The size of it is 512 × 614 × 176, with a spatial resolution of 18 m. After removing water absorption and low SNR bands, 176 bands were used for the analysis.

TABLE I
Land Cover Classes and the Numbers of Samples in the Chikusei Dataset

ID	land-covers	samples
1	Water	2845
2	Bare soil (school)	2859
3	Bare soil (park)	286
4	Bare soil (farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forests	20516
8	Grass	6515
9	Rice field (grown)	13369
10	Rice field (first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (non-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	Total	**77592**

Regarding the performance evaluation, we adopt the class-specified accuracy (CA), average accuracy (AA), overall accuracy (OA), and Kappa coefficient (Kappa) as criteria, which are commonly used in HSIC tasks.

B. Experimental Settings

To maximize the effectiveness of the proposed networks, we conduct thorough experimental configurations. For the meta-training process, the mini-batch, or episodes, constructed from both the source and target datasets, is iteratively input into the networks. The total number of the training episodes is experimentally set to 10 000, which proved to be adequate for network training. The learning rate is set to 0.001, following a standard setup. Additionally, the numbers of the support samples N_S and query samples N_Q are set to 1 and 19, respectively. The patch size is set to 9 × 9 in each episode, just the same as the previous few-shot HSIC works [42], [43]. Moreover, the sampled classes are set to 16 for IP and HC, 9 for UP, and 13 for KSC dataset. This configuration aims to facilitate collaborative learning of a shared discriminant embedding space for both source and target classes.

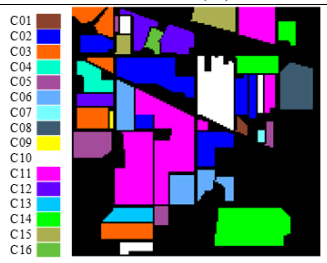
As to the MSFormer, we divide each patch into nine submaps of uniform shape. Additionally, the kernel size of the DWC used to squeeze the spatial dimension is 7 × 7. Besides, we test the appropriate length of feature vectors extracted by

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TABLE II

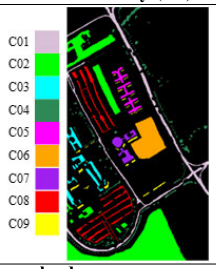
GROUNDTRUTH AND THE DETAILED NUMBERS OF THE SAMPLES FOR EACH CLASS IN THE TARGET DATASETS

Indian Pines (IP)



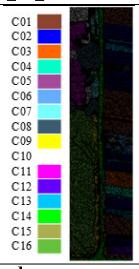
ID	land-covers	samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
	Total	**10249**

Pavia University (UP)



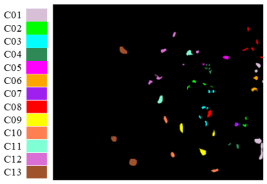
ID	land-covers	samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947
	Total	**42776**

WHU_Hi_HanChuan (HC)



ID	land-covers	samples
1	Strawberry	6710
2	Cowpea	3412
3	Soybean	1543
4	Sorghum	802
5	Water spinach	180
6	Watermelon	679
7	Greens	885
8	Trees	2696
9	Grass	1420
10	Red roof	1577
11	Gray roof	2536
12	Plastic	551
13	Bare soil	1367
14	Road	2784
15	Bright object	170
16	Water	11310
	Total	**38622**

Kennedy Space Center (KSC)



ID	land-covers	samples
1	Scrub	761
2	Willow swamp	243
3	CP hammock	256
4	Slash pine	252
5	Oak/Broadleaf	161
6	Hardwood	229
7	Swap	105
8	Graminoid marsh	431
9	Spartina marsh	520
10	Cattail marsh	404
11	Salt marsh	419
12	Mud flat	503
13	Water	927
	Total	**5211**

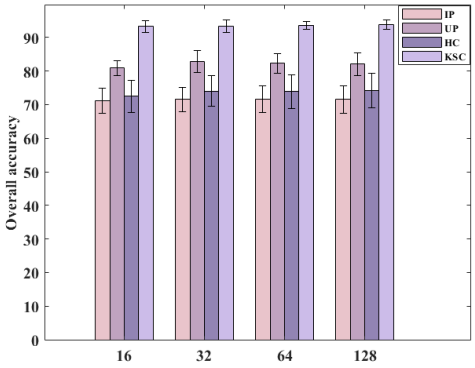


Fig. 7. OA with standard deviation on IP, UP, HC, and KSC for different feature dimensions.

the MSFormer. As illustrated in Fig. 7, the highest and most stable OAs occur when the feature dimension is set to 32 for IP, HC, and UP and set to 128 for KSC.

Moreover, we conduct comprehensive experiments to assess the impact of hyperparameters on the three test datasets. Specifically, we focus on the weight hyperparameter λ, temperature factor τ associated with the supervised contrastive loss, and temperature factor ϵ linked to the unsupervised contrastive loss. The λ is chosen as 1e-4 and 1e-5, and we explore the mutual influence of τ and ϵ on performance. As shown in Figs. 8 and 9, the optimal results are obtained when λ, τ, and ϵ are set to (1e-5, 0.5, 0.01), (1e-5, 0.1, 0.01), (1e-4, 0.01, 0.05), and (1e-4, 0.5, 0.05) for the IP, UP, HC, and KSC datasets, respectively. From these experiments, it is evident that the outcomes are notably influenced by these three hyperparameters, with particular sensitivity observed for the weight factor λ, which plays a crucial role in balancing the aggregation of similar data and the separation of dissimilar data.

TABLE III

CLASSIFICATION PERFORMANCE (%) OF THE ABLATIONS ON THE PROPOSED METHOD ON IP, UP, HC, AND KSC

Data set	Case	F	L_{sc}	L_{uc}	AA	OA	Kappa
IP	1	✗	✓	✓	75.77 ± 1.16	64.12 ± 2.46	59.59 ± 2.60
	2	✓	✗	✓	81.00 ± 1.31	70.43 ± 2.97	66.66 ± 3.13
	3	✓	✓	✗	81.14 ± 1.64	70.14 ± 3.22	66.38 ± 3.45
	4	✓	✓	✓	**82.23 ± 1.75**	**71.51 ± 3.69**	**67.90 ± 3.98**
UP	1	✗	✓	✓	81.32 ± 1.91	79.39 ± 4.13	73.57 ± 4.77
	2	✓	✗	✓	85.06 ± 2.20	82.46 ± 3.55	77.43 ± 4.16
	3	✓	✓	✗	83.83 ± 2.41	81.54 ± 3.26	76.21 ± 3.66
	4	✓	✓	✓	**85.67 ± 1.92**	**82.90 ± 3.26**	**77.99 ± 3.82**
HC	1	✗	✓	✓	60.67 ± 2.22	67.46 ± 4.07	62.64 ± 4.34
	2	✓	✗	✓	69.92 ± 2.58	73.06 ± 4.67	69.07 ± 5.09
	3	✓	✓	✗	70.44 ± 2.46	73.43 ± 4.78	69.50 ± 5.19
	4	✓	✓	✓	**70.60 ± 2.80**	**74.02 ± 4.54**	**70.14 ± 4.93**
KSC	1	✗	✓	✓	85.76 ± 1.71	89.14 ± 1.49	87.92 ± 1.66
	2	✓	✗	✓	91.21 ± 1.71	93.36 ± 1.50	92.60 ± 1.67
	3	✓	✓	✗	91.87 ± 2.01	93.87 ± 1.68	93.18 ± 1.87
	4	✓	✓	✓	**92.12 ± 1.51**	**94.17 ± 1.23**	**93.51 ± 1.37**

C. Ablations on the Method

As described in Section III, the proposed method mainly comprises the CNN-Transformer network for feature extraction and a hybrid loss function that combines the classification loss, supervised contrastive loss, and unsupervised contrastive loss. To verify the effectiveness of each module, we conduct ablation studies focusing primarily on these components.

Specifically, we compare the backbone F composed of LSSICM and MSFormer with a 3D-ResNet normal feature extractor [43]. Then, we employ L_{cls} in each case and compare the overall structure with partial structures. The four cases are detailed as follows.

Case 1: The proposed network with L_{sc} and L_{uc}, utilizing 3D-ResNet as the feature extractor.

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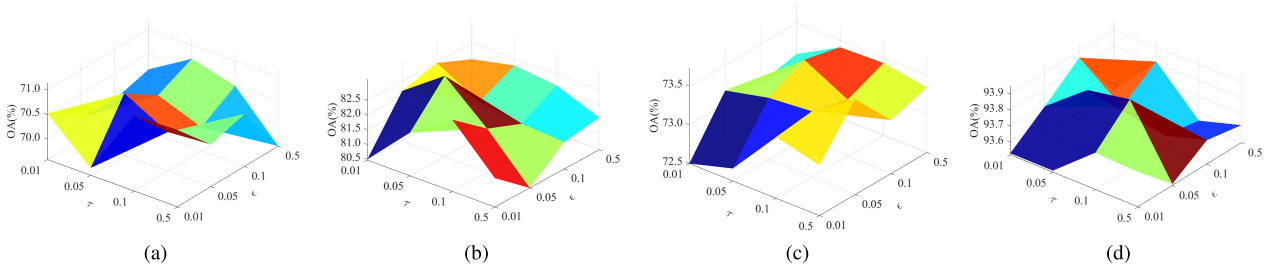


Fig. 8. Parameter tuning of τ and ϵ when λ is 1e-5 in the four datasets. (a) IP. (b) UP. (c) HC. (d) KSC.

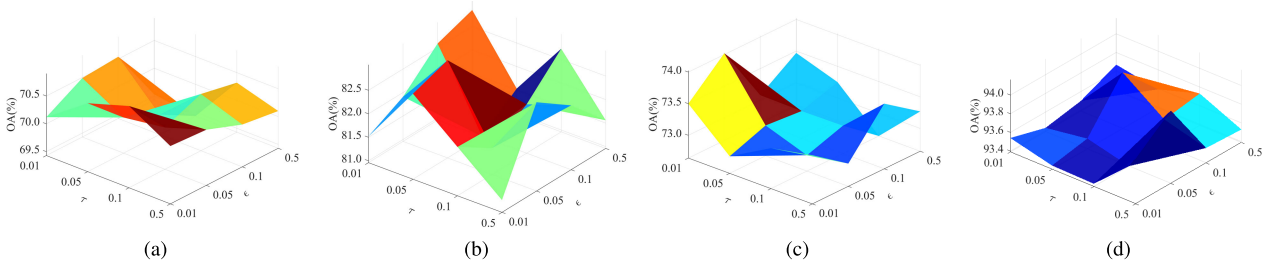


Fig. 9. Parameter tuning of τ and ϵ when λ is 1e-4 in the four datasets. (a) IP. (b) UP. (c) HC. (d) KSC.

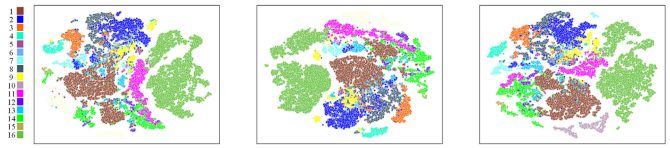


Fig. 10. Visualization of embedded feature vectors by t-SNE maps on IP dataset in different cases.



Fig. 11. Visualization of embedded feature vectors by t-SNE maps on UP dataset in different cases.

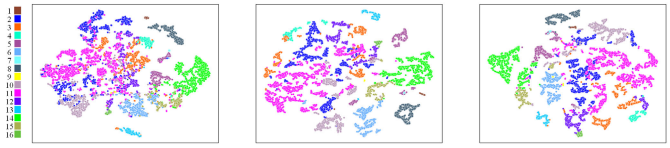


Fig. 12. Visualization of embedded feature vectors by t-SNE maps on HC dataset in different cases.

Case 2: The proposed network with backbone F and L_{uc}.

Case 3: The proposed network with backbone F and L_{sc}.

Case 4: The proposed CTF-SSCL with backbone F, L_{sc}, and L_{uc}.

We randomly select five labeled samples in each novel class (i.e., five shots), and the experiments are repeated ten times independently. The average classification performance and the standard deviation on the three experiment datasets are presented in Table III. Specifically, the best and second-best

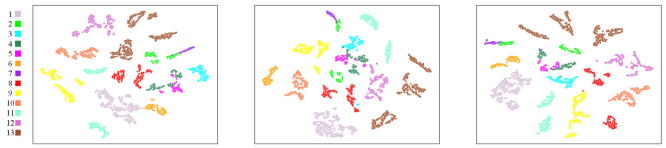


Fig. 13. Visualization of embedded feature vectors by t-SNE maps on KSC dataset in different cases.

results in OA, AA, and Kappa are highlighted in bold and underlined, respectively. From Table III, it is evident that our proposed backbone F, L_{sc}, and L_{uc} can significantly contribute to the improvement of OA, AA, and Kappa. Specifically, LSSICM facilitates the interaction of spectral–spatial information, while MSFormer effectively extracts both local and global features for HSIC. Furthermore, the hybrid loss function proves to be crucial in optimizing the feature space by consolidating labeled samples of the same category and capturing similarities among different views of unlabeled samples. Moreover, we visually inspect the feature vectors extracted under cases 2–4 using the t-distributed stochastic neighbor embedding (t-SNE) method. Figs. 10–13 show the feature distribution of IP, UP, HC, and KSC datasets, respectively. Take UP as an example, it can be observed that the feature vectors from class 7 (depicted in purple) in case 4 exhibit higher concentration and less overlap with other classes compared to cases 2 and 3 in Fig. 11. This observation underscores the significant role of semisupervised contrastive loss in extracting discriminative features.

D. Comparing With Other Methods

To validate the effectiveness of the proposed CTF-SSCL, we conduct a comparative analysis with other state-of-the-art methods. Specifically, we compare its classification outcomes with well-known DL-based spectral–spatial classifiers, namely,

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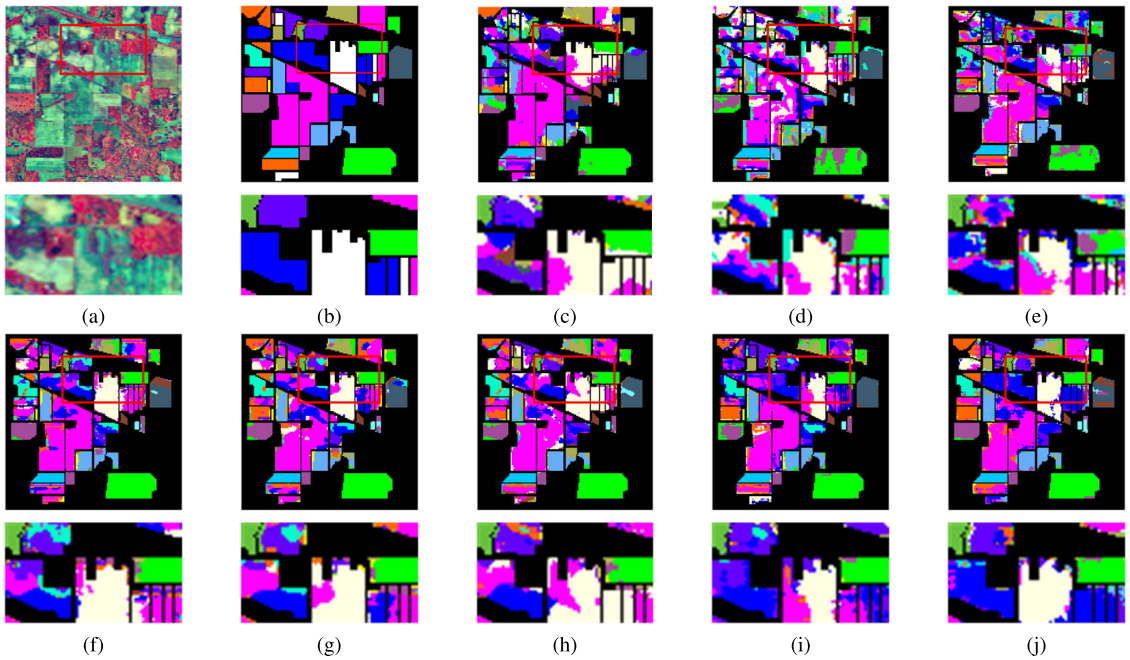


Fig. 14. Classification maps and their partially enlarged views obtained via various algorithms for IP dataset. (a) False-color image. (b) Groundtruth. (c) 3D-CNN. (d) SpectralFormer. (e) DFSL. (f) DCFSL. (g) CMFSL. (h) Gia-CFSL. (i) ADAFSL. (j) CTF-SSCL. Best zoomed-in view.

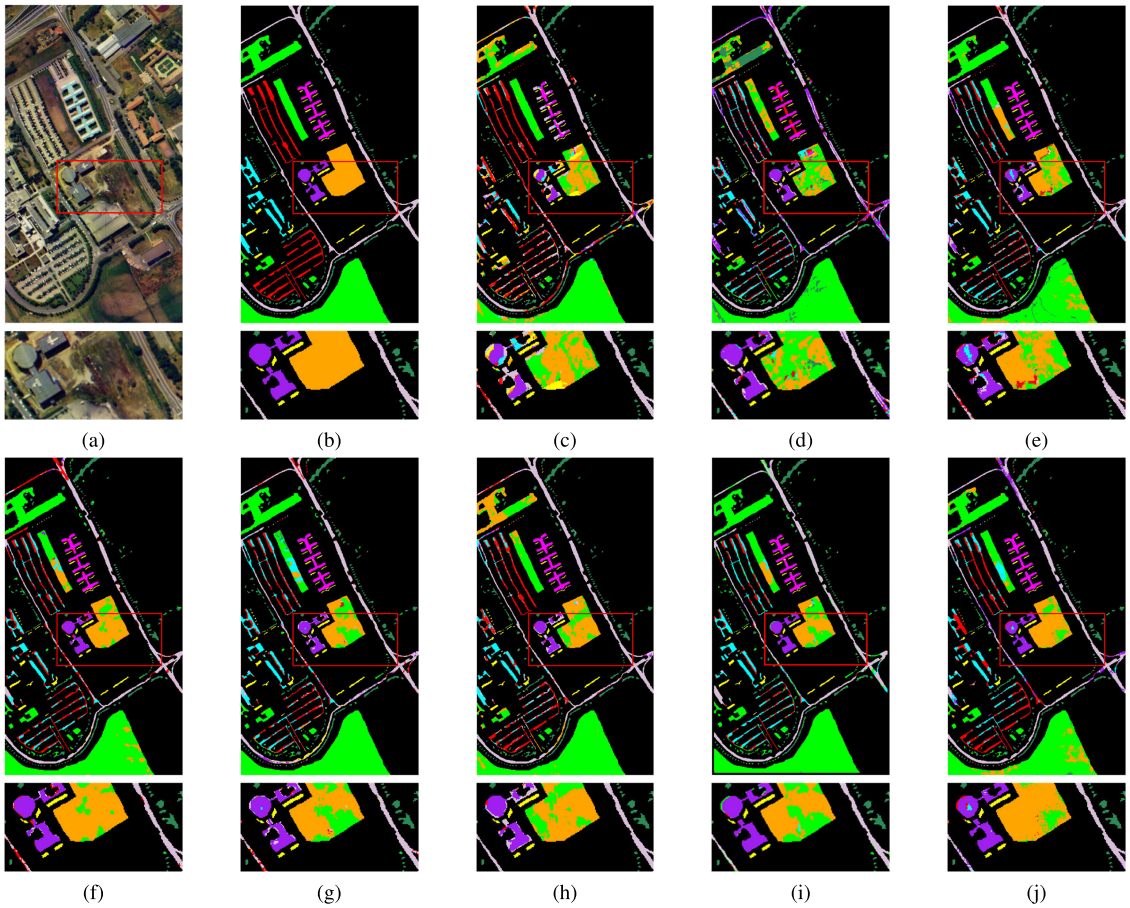


Fig. 15. Classification maps and their partially enlarged views obtained via various algorithms for UP dataset. (a) False-color image. (b) Groundtruth. (c) 3D-CNN. (d) SpectralFormer. (e) DFSL. (f) DCFSL. (g) CMFSL. (h) Gia-CFSL. (i) ADAFSL. (j) CTF-SSCL. Best zoomed-in view.

3D-CNN [28] and SpectralFormer [37], representing CNN and transformer methods, respectively. Additionally, we evaluate the performance against DFSL [42], which was the pioneering work to introduce the FSL paradigm to HSIC.

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TABLE IV
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE IP DATASET

No.	3D-CNN [28]	SpectralFormer [37]	DFSL [42]	DCFSL [43]	CMFSL [56]	Gia-CFSL [57]	ADAFSL [58]	CTF-SSCL
1	96.04±6.77	69.27±13.23	93.17±3.75	95.61±8.64	**99.51±0.98**	90.00±12.36	97.56±3.27	98.05±2.39
2	30.38±21.81	18.74±10.16	36.33±7.74	44.88±11.94	45.29±6.81	49.13±7.14	**56.90±12.39**	50.74±15.76
3	35.99±13.22	22.84±11.97	36.07±7.00	45.64±7.81	54.55±7.02	51.21±7.77	**63.30±6.48**	61.15±11.29
4	55.07±16.04	25.00±9.64	53.79±12.68	74.91±11.26	78.49±14.06	75.73±12.97	82.33±15.58	**87.03±12.90**
5	41.08±19.38	41.11±14.98	64.39±6.19	73.53±8.48	77.14±7.21	73.62±7.16	78.47±7.99	**78.49±8.50**
6	62.31±20.74	38.73±10.81	73.69±7.61	85.16±5.62	84.94±8.22	83.37±6.98	86.66±7.74	**89.06±9.04**
7	**100.00±0.00**	75.22±17.93	98.69±2.79	99.56±1.30	99.56±1.30	97.83±4.01	99.56±1.30	**100.00±0.00**
8	90.86±7.48	82.92±5.41	73.70±14.51	82.09±13.14	90.38±9.86	87.10±11.52	89.96±7.85	**92.73±10.80**
9	95.83±6.11	85.33±19.51	98.00±4.27	99.33±2.00	99.33±2.00	98.00±3.06	**100.00±0.00**	98.00±3.06
10	35.48±23.34	24.45±10.62	54.19±8.85	61.40±8.65	59.53±6.02	59.74±9.92	69.74±7.71	**69.96±10.18**
11	51.40±20.02	35.59±17.26	54.06±13.77	59.98±11.42	53.41±13.15	59.08±10.27	57.82±13.69	**64.04±14.99**
12	37.93±13.67	22.45±3.88	29.05±7.82	41.96±12.29	50.17±14.68	46.84±14.42	**60.68±13.23**	57.24±12.90
13	87.50±13.83	72.65±12.31	96.00±4.80	99.20±0.78	98.20±3.78	98.15±4.28	97.95±1.33	**99.50±0.92**
14	79.51±7.79	47.04±24.44	79.06±8.61	84.01±9.80	83.11±9.22	77.24±10.56	85.80±9.93	**89.50±6.26**
15	65.26±12.28	29.74±9.48	52.28±15.16	68.14±9.01	79.24±15.29	67.61±8.71	76.33±11.17	**81.65±8.76**
16	88.92±9.60	64.66±20.87	95.80±6.55	98.64±2.26	**99.66±0.52**	99.09±1.22	98.18±2.28	98.52±2.79
AA	65.85±2.28	47.23±3.01	68.02±1.86	75.88±1.28	78.28±0.62	75.86±1.58	81.33±2.31	**82.23±1.75**
OA	52.52±4.25	35.30±3.47	56.02±2.20	64.28±2.43	64.70±2.58	64.46±2.35	70.13±3.87	**71.51±3.69**
Kappa	46.68±4.40	27.91±3.20	50.67±2.14	59.76±2.59	60.24±2.62	61.84±3.13	66.44±4.12	**67.90±3.98**

TABLE V
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE UP DATASET

No.	3D-CNN [28]	SpectralFormer [37]	DFSL [42]	DCFSL [43]	CMFSL [56]	Gia-CFSL [57]	ADAFSL [58]	CTF-SSCL
1	56.31±16.91	46.71±11.59	78.65±7.81	74.69±17.36	78.67±8.95	82.36±7.44	**86.93±7.07**	85.97±4.58
2	80.18±12.07	77.85±6.74	76.31±6.75	**87.81±7.08**	78.51±11.70	85.67±5.58	79.13±11.15	82.11±6.98
3	38.91±12.21	50.20±10.16	52.27±8.66	62.99±13.31	65.32±11.62	57.91±12.44	**70.49±5.51**	69.78±13.91
4	84.17±12.28	74.31±10.01	88.16±8.27	91.66±5.00	91.78±4.27	**92.26±3.86**	88.87±6.05	90.91±3.13
5	82.04±12.24	93.07±7.26	98.80±1.00	98.14±3.72	98.84±1.97	99.53±0.40	97.69±3.33	**99.75±0.37**
6	47.27±15.36	42.47±7.85	60.50±9.72	71.03±8.36	**80.80±12.27**	73.33±8.53	79.73±9.60	73.65±9.62
7	48.61±14.16	73.47±9.40	65.09±8.27	78.82±8.18	81.46±7.06	78.94±7.90	89.37±5.73	**91.52±5.65**
8	43.01±17.75	43.43±13.98	59.56±7.12	64.93±12.23	75.24±9.32	74.18±11.87	80.58±11.27	**82.50±9.69**
9	88.25±9.83	95.37±4.84	98.47±1.58	99.11±0.84	**99.44±0.98**	98.59±1.79	96.96±5.16	94.79±6.02
AA	63.19±3.32	66.32±1.92	75.31±1.91	81.02±2.58	83.34±1.31	82.53±1.90	85.53±1.40	**85.67±1.92**
OA	66.93±4.50	65.02±3.13	73.89±2.97	81.19±3.40	80.01±4.11	82.34±2.35	82.10±4.05	**82.90±3.26**
Kappa	56.96±5.12	54.76±3.39	66.49±3.34	75.53±4.10	74.57±4.60	77.09±2.83	77.11±4.56	**77.99±3.82**

TABLE VI
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE HC DATASET

No.	3D-CNN [28]	SpectralFormer [37]	DFSL [42]	DCFSL [43]	CMFSL [56]	Gia-CFSL [57]	ADAFSL [58]	CTF-SSCL
1	43.32±13.53	48.87±8.48	73.49±9.21	74.27±9.81	65.03±7.80	74.60±11.42	71.48±11.91	**78.99±7.89**
2	38.20±12.86	31.36±5.96	38.89±5.78	47.44±9.37	**55.82±7.98**	39.13±9.66	38.75±8.71	50.20±4.59
3	23.87±11.35	25.22±10.05	45.86±6.87	56.58±13.79	60.93±3.77	61.20±15.39	74.15±10.01	**76.46±11.96**
4	71.57±20.27	58.02±22.60	79.20±16.48	90.73±5.19	90.43±5.83	89.07±6.01	95.12±2.45	**96.18±1.35**
5	67.26±18.55	68.34±13.75	86.51±7.64	87.43±6.76	87.53±7.88	84.06±8.17	89.03±9.10	**93.94±5.08**
6	13.47±6.12	16.11±4.62	23.01±5.83	26.32±9.05	34.24±8.66	35.36±12.19	42.55±10.82	**49.10±10.21**
7	71.37±16.11	75.63±19.37	73.91±12.91	84.06±10.85	**85.09±9.61**	72.59±12.11	82.23±10.23	81.69±10.29
8	21.25±14.49	27.75±10.55	32.17±10.57	34.75±8.16	45.00±7.56	37.43±12.89	36.29±13.19	**49.73±11.44**
9	17.19±8.28	21.80±8.93	32.78±7.86	32.59±9.03	43.19±13.78	35.18±12.28	43.96±13.67	**48.19±13.62**
10	22.73±9.53	53.78±13.87	67.84±12.03	69.64±9.65	68.37±12.61	71.27±16.98	78.05±14.77	**86.18±13.61**
11	30.73±19.13	64.21±15.66	60.47±8.13	68.82±11.00	66.25±9.94	67.65±9.67	74.49±8.41	**79.74±7.54**
12	35.86±13.17	37.51±7.51	48.75±9.75	45.92±5.61	49.78±17.05	45.20±10.72	48.88±15.19	**53.37±9.62**
13	37.24±9.26	31.07±10.19	39.53±6.54	40.05±8.21	43.58±7.94	43.01±7.88	45.15±7.64	**50.26±8.25**
14	41.01±12.49	52.36±10.36	57.94±11.97	62.59±9.84	61.79±12.73	60.00±9.02	**66.21±13.81**	63.19±12.80
15	55.09±16.17	63.76±6.49	66.06±11.83	69.27±12.05	79.15±10.20	68.91±9.16	48.30±11.62	**82.91±3.05**
16	83.59±9.02	71.97±8.56	86.01±10.81	87.62±10.23	85.02±10.48	89.10±11.48	87.95±11.65	**89.37±10.34**
AA	42.11±4.01	46.74±3.06	57.03±1.48	61.13±3.16	63.83±2.71	60.86±2.91	63.91±2.37	**70.60±2.80**
OA	50.11±2.06	51.73±2.14	64.13±3.73	67.58±3.62	67.33±4.28	67.55±4.01	69.10±4.32	**74.02±4.54**
Kappa	42.70±2.95	45.61±2.20	58.92±3.83	62.74±3.90	62.64±4.59	62.73±4.28	64.62±4.58	**70.14±4.93**

Furthermore, recent advancements in addressing the few-shot problem for HSIC have led to the development of DCFSL [43], CMFSL [56], Gia-CFSL [57], and ADAFSL [58], which have demonstrated superior performance. Hence, we include DFSL, DCFSL, CMFSL, Gia-CFSL, and ADAFSL as comparative methods in our evaluation. The Chikusei dataset is utilized as

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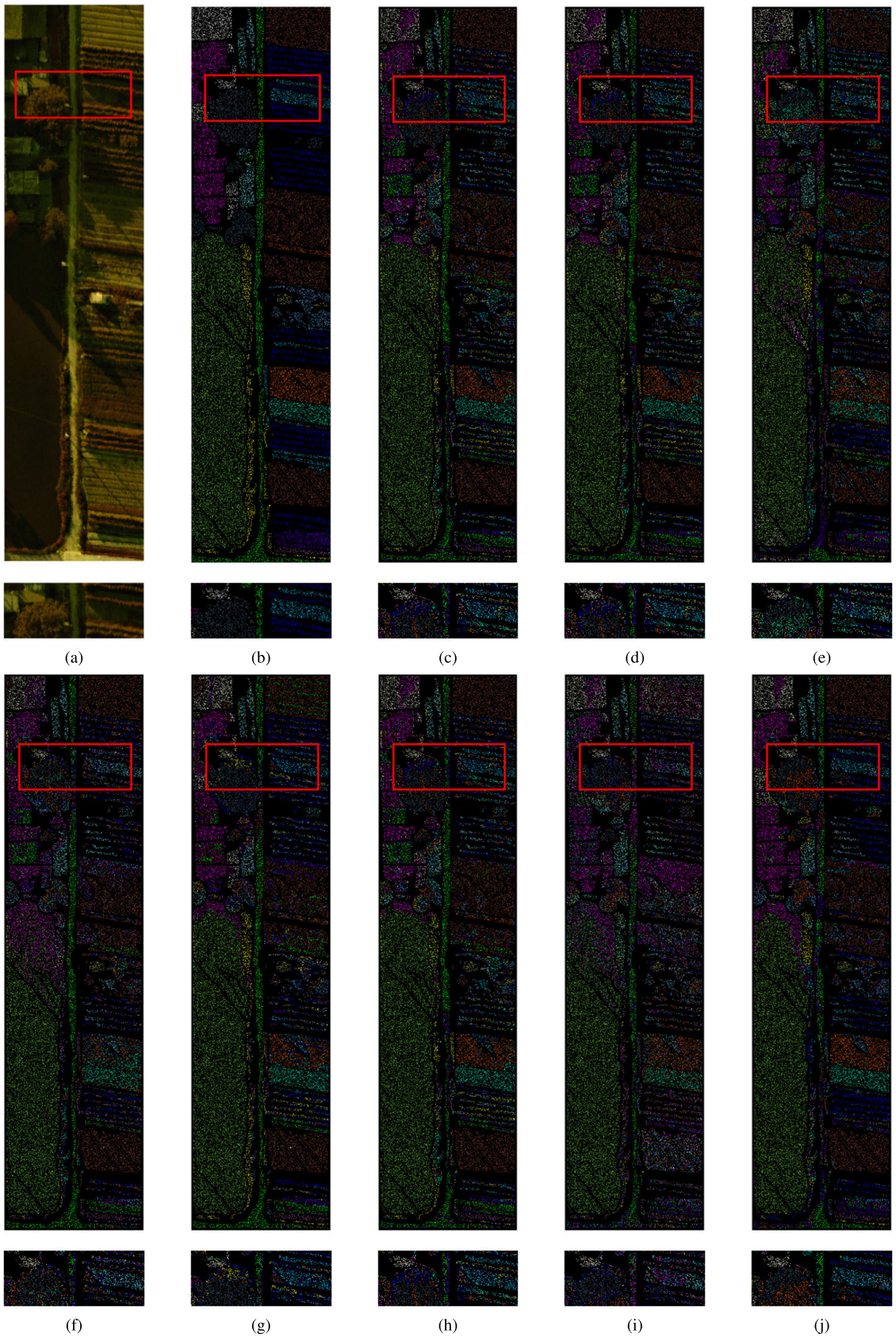


Fig. 16. Classification maps and their partially enlarged views obtained via various algorithms for HC dataset. (a) False-color image. (b) Groundtruth. (c) 3D-CNN. (d) SpectralFormer. (e) DFSL. (f) DCFSL. (g) CMFSL. (h) Gia-CFSL. (i) ADAFSL. (j) CTF-SSCL. Best zoomed-in view.

the source dataset for these FSL methods, and the hyperparameter configurations are set according to the original articles. Five samples are randomly selected from each target class for inclusion in the training process. To ensure the robustness

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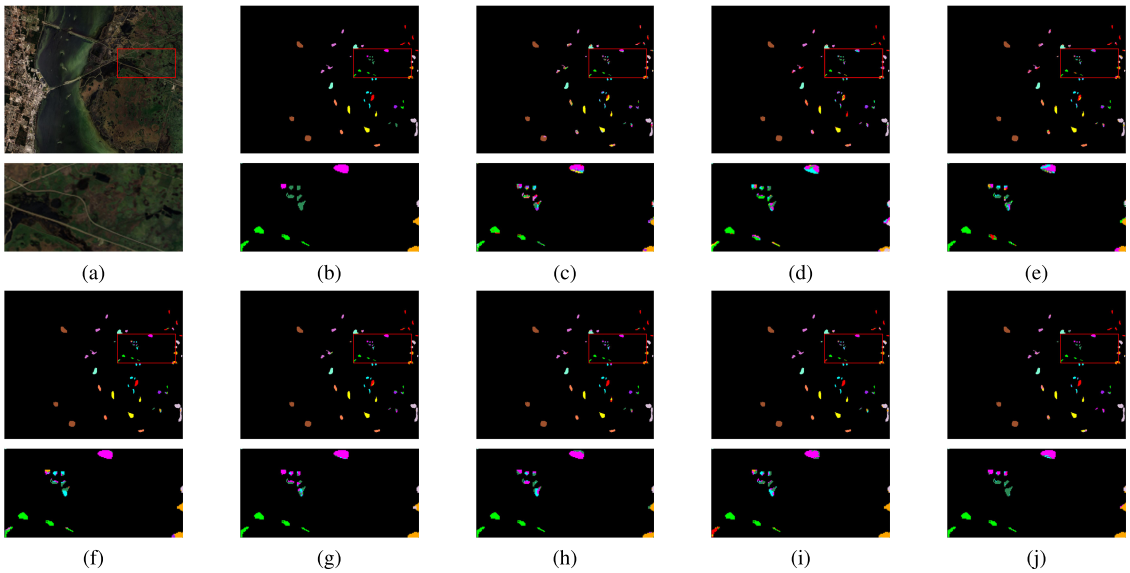


Fig. 17. Classification maps and their partially enlarged views obtained via various algorithms for KSC dataset. (a) False-color image. (b) Groundtruth. (c) 3D-CNN. (d) SpectralFormer. (e) DFSL. (f) DCFSL. (g) CMFSL. (h) Gia-CFSL. (i) ADAFSL. (j) CTF-SSCL. Best zoomed-in view.

of the experimental results, each method and dataset undergo ten independent runs. The mean accuracies and the standard deviations are presented from Tables IV to VII. To enhance clarity, the highest and suboptimum accuracies in each row are highlighted in bold and underlined, respectively, for easier comparison and interpretation of the results.

Table IV demonstrates the classification performance across various methods on the IP dataset. Notably, CTF-SSCL outperforms other comparisons in terms of AA, OA, and Kappa. In ten categories, CTF-SSCL secures the first position and in four categories, it ranks second. The AA, OA, and Kappa surmount the second-best values by 0.90%, 1.38%, and 1.46%, respectively. This result indicates the capability of our method to extract more discriminative information from the original data, making it easier to distinguish between different data features. Moving on to Table V which provides the classification results on the UP dataset, it can be observed that the presented CTF-SSCL achieves outstanding classification performance compared to other methods. Particularly, CTF-SSCL secures the first and second positions in three and two categories, respectively, showcasing its effectiveness in this dataset. Furthermore, Table VI summarizes the classification accuracies on the HC datasets. The proposed method achieves the top-level classification performance, again demonstrating the capacity of the CTF-SSCL frameworks. It is noteworthy that the AA, OA, and Kappa metrics exceed the second-best values by 6.69%, 4.92%, and 5.52%, respectively, underscoring the effectiveness of the CTF-SSCL frameworks. As shown in Table VII for KSC, CTF-SSCL is better than other comparisons in terms of AA, OA, and Kappa. In eight categories, CTF-SSCL achieves the top position, while in three categories, it attains the second place, demonstrating exceptional performance. Additionally, the AA, OA, and Kappa exceed the second-best values by 2.19%, 1.07%, and 1.19%, respectively.

In general, both CNN-based and transformer-based methods encounter performance challenges when dealing with limited labeled samples. Methods incorporating the FSL paradigm perform better in AA, OA, and Kappa than those without FSL, validating the potential of the FSL for HSIC. In particular, the CTF-SSCL achieves the highest performance in all four datasets, highlighting its efficacy in HSIC tasks.

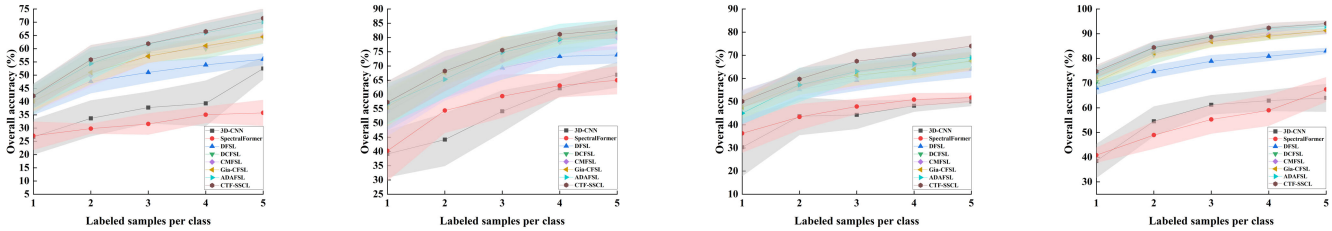
To facilitate visual comparison, we present the optimal classification maps generated by various methods on the three datasets from Figs. 14–17, aligning with the data presented in Tables IV to VII. Figs. 14–17 include the corresponding composite false-color images and groundtruth for comprehensive evaluation. Additionally, a local magnification of the red box in each classification map enhances the clarity of the comparison. In Fig. 14, it is evident that the classification maps produced by different methods for the IP dataset contain numerous undesired misclassifications attributed to noise and intractable spectral uncertainty issue within the dataset. Thus, it is challenging to facilitate the methods to produce a satisfactory classification map with limited few-shot labeled samples. By contrast, in Fig. 15 for the UP dataset, the classification map generated by CTF-SSCL closely aligns with the groundtruth and exhibits minimal noise. This phenomenon demonstrates the ability of the approach to extract distinctive interclass features, which is again verified in Figs. 16 and 17. However, due to the presence of multiple shadow coverage areas, it is pretty hard to clearly illustrate the classification maps of the HC.

Furthermore, to assess the effectiveness of the proposed method under reduced labeled samples in the target classes, we randomly select annotated samples ranging from 1 to 5 and conduct the experiments ten times. The mean OAs of various methods on the three datasets are illustrated in Fig. 18. Remarkably, CTF-SSCL ranks first place in all scenarios, affirming the robustness of the proposed networks. Additionally, the FSL-based approaches exhibit more robust generalization capabilities compared to other methods due to the utilization of the meta-learning strategy.

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TABLE VII
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE KSC DATASET

No.	3D-CNN [28]	SpectralFormer [37]	DFSL [42]	DCFSL [43]	CMFSL [56]	Gia-CFSL [57]	ADAFSL [58]	CTF-SSCL
1	75.69±13.65	76.25±25.25	94.52±5.71	97.42±1.47	92.26±3.26	95.54±3.83	95.90±2.84	**97.55±1.62**
2	36.34±14.04	40.93±13.03	71.26±11.26	83.40±8.42	90.17±10.15	83.11±11.64	88.61±8.39	**94.20±7.40**
3	48.37±14.55	37.01±22.36	74.50±9.93	85.94±12.28	86.81±10.38	86.02±10.02	**92.59±5.63**	90.16±10.39
4	20.32±5.85	23.64±8.23	39.31±10.46	48.14±16.86	52.43±11.38	52.55±13.19	49.88±14.45	**65.71±16.79**
5	40.77±17.58	21.47±12.43	62.57±10.29	71.15±9.14	81.73±7.44	76.22±8.06	76.47±9.64	**84.68±6.22**
6	53.44±16.00	38.53±17.98	81.52±5.79	90.71±4.38	92.01±8.17	89.38±7.12	92.95±11.63	**95.13±6.73**
7	81.90±13.15	69.00±18.29	97.40±4.43	96.70±6.28	92.50±11.24	**99.00±2.41**	93.30±11.53	96.70±6.78
8	48.48±13.98	38.05±12.81	69.74±9.71	85.99±11.30	85.10±11.23	88.24±8.06	**95.45±4.51**	95.05±5.56
9	79.71±9.73	63.32±15.48	85.67±13.88	90.46±10.09	90.68±9.53	92.16±8.69	**95.22±6.81**	92.31±8.83
10	45.66±12.42	70.43±9.14	75.36±12.63	97.82±1.39	97.09±4.19	96.02±2.40	**97.89±2.42**	92.23±5.76
11	84.37±7.04	89.76±8.33	96.71±2.40	95.46±5.47	96.59±3.21	95.58±5.28	96.33±3.66	**97.61±2.19**
12	43.64±8.40	78.82±7.46	73.94±9.14	93.86±4.36	92.73±5.88	92.61±5.62	94.44±2.67	**96.26±3.83**
13	90.43±19.78	**100.00±0.00**	99.69±0.58	**100.00±0.00**	99.97±0.10	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
AA	57.62±4.81	57.48±4.35	78.63±1.58	87.47±1.75	88.47±1.36	88.18±1.17	89.93±1.79	**92.12±1.51**
OA	64.03±5.67	67.42±5.02	83.03±1.91	91.09±1.27	91.06±1.34	91.26±1.11	93.10±1.19	**94.17±1.23**
Kappa	60.09±6.11	63.75±5.46	81.09±2.12	90.08±1.41	90.06±1.48	90.28±1.23	92.32±1.33	**93.51±1.37**



(a) (b) (c) (d)

Fig. 18. OA (%) of various methods on the four datasets with different numbers of labeled samples. (a) IP. (b) UP. (c) HC. (d) KSC.

TABLE VIII
PARAMETERS AND TRAINING TIME OF THE PROPOSED METHOD AND THE COMPETITIVE FSL-BASED METHODS

		IP	UP	HC	KSC
DCFSL	para.	70,696	59,869	78,096	67,813
	training time(s)	1345.29	839.40	1433.90	1108.01
CMFSL	para.	1,996,662	1,406,838	2,457,462	1,849,206
	training time(s)	1422.86	768.04	1566.46	1329.40
Gia-CFSL	para.	218,392	208,692	225,792	215,992
	training time(s)	3458.83	2530.66	3553.73	3551.67
ADAFSL	para.	60,064	49,237	67,464	57,181
	training time(s)	26443.83	37523.96	39826.56	31793.93
CTF-SSCL	para.	397,968	390,357	407,688	411,421
	training time(s)	1200.57	746.10	1304.12	1079.92

E. Analysis of the Computational Complexity

In this section, we delve into the computational complexity of the CTF-SSCL. Specifically, we conduct a comparative analysis focusing on the number of parameters (para.) and the training time of the proposed CTF-SSCL and the competitive few-shot HSIC methods, including DCFSL, CMFSL, Gia-CFSL, and ADAFSL. All procedures are performed on the NVIDIA GeForce GTX 4090Ti GPU, with detailed statistics summarized in Table VIII.

From Table VIII, it can be observed that the CTF-SSCL has a moderate number of parameters compared to alternative methods, which enables the model to effectively transfer knowledge from the source to the target classes, thereby alleviating the risk of overfitting. Impressively, the training time of CTF-SSCL across the four datasets is notably shorter. This suggests that despite the substantial memory requirements stemming from the parameter count, the computational burden is relatively light, resulting in expedited training time. Particularly, due to the fewer classes in comparison to IP, HC, and KSC, the training time of the UP within our method is considerably shorter than that of the other three datasets.

V. CONCLUSION

In this article, we propose a novel CNN-Transformer network for few-shot HSIC with semisupervised contrastive learning to improve the classification performance under the scarcity of labeled samples. Initially, an LSSICM comprised of the GWC and CCFM is proposed to extract local spectral–spatial features in an efficient manner, addressing concerns related to excessive parameters. Subsequently, the MSFormer is introduced to aggregate information across various scales, facilitating the capture of both local and global features effectively. To further enhance feature extraction, a semisupervised loss function is formulated, incorporating both supervised and unsupervised CL components to encourage the network to learn more discriminative features in the latent space. Additionally, we employ the SFSS to generate sample pairs for unsupervised contrastive loss computation, tailored specifically for HSI data. Extensive experimental evaluations confirm the efficacy of our proposed CTF-SSCL in dealing with the few-shot HSIC tasks.

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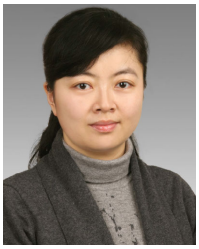
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